



GETTING STARTED

1. Install

In the terminal
`sudo pip install plotly`

2. Sign Up & Configure

<http://www.plot.ly/python/getting-started>

3. Boilerplate Imports

```
import plotly.plotly as py
import plotly.graph_objs as go
```

4. A Hello World Figure

```
trace = {'x': [1, 2], 'y': [1, 2]}
data = [trace]
data = {}
fig = go.Figure (
    data = data, layout = layout )
```

5. Plot the Figure!

In the terminal:
`plot_url = py.plot ( fig )`

Or in the IPython notebook:
`py.iplot ( fig )`

BASIC CHARTS

Line Plots

```
trace1 = go.Scatter (
    x = [1, 2], y = [1, 2])
trace2 = go.Scatter (
    x = [1, 2], y = [2, 1])
py.iplot ( [ trace1, trace2 ] )
```

Bubble Charts

```
trace = go.Scatter (
    x = [1, 2, 3], y = [1, 2, 3],
    marker = dict (
        color = ['red', 'blue',
        'green']
        size = [30, 80, 200] ),
    mode = 'markers' )
py.iplot ( [ trace ] )
```

Scatter Plots

```
trace1 = go.Scatter (
    x = [1, 2, 3], y = [1, 2, 3],
    text = ['A', 'B', 'C'],
    textposition = 'top center'
    mode = 'markers+text' )
mode = [ trace ]
py.iplot ( data )
```

Heatmaps

```
trace = go.Heatmap (
    z = [[1, 2, 3, 4],
    [5, 6, 7, 8]])
data = [ trace ]
py.iplot ( data )
```

Bar Charts

```
trace = go.Bar (
    x = [1, 2], y = [1, 2])
data = [ trace ]
py.iplot ( data )
```

Area Plots

```
trace = go.Scatter (
    x = [1, 2], y = [1, 2],
    fill = 'tonexty' )
data = [ trace ]
py.iplot ( data )
```

LAYOUT

Legends

```
trace1 = go.Scatter (
    name = 'Calvin'
    x = [1, 2], y = [1, 2])
```

```
trace2 = go.Scatter (
    name = 'Hobbes'
    x = [2, 1], y = [2, 1])
```

```
layout = go.Layout (
    showlegend = True,
    legend = dict (
        x = 0.2, y = 0.5 )
    )
```

```
data = [ trace1, trace2 ]
fig = go.Figure (
    data = data,
    layout = layout )
py.iplot ( fig )
```

Axes

```
trace = go.Scatter (
    x = [1, 2, 3, 4],
    y = [1, 2, 3, 6])
```

```
axis_template = dict (
    showgrid = False,
    zeroline = False,
    nticks = 20,
    showline = True,
    title = 'X AXIS'
    mirror = 'all' )
layout = go.Layout (
    xaxis = axis_template,
    yaxis = axis_template,
    )
```

```
data = [ trace ]
fig = go.Figure (
    data = data
    layout = layout )
py.iplot ( fig )
```

STATISTICAL CHARTS

Histograms

```
trace = go.Histogram (
    x = [ 1, 2, 3, 3, 3, 4, 5 ] )
data = [ trace ]
py.iplot ( data )
```

Box Plots

```
trace = go.Box (
    x = [ 1, 2, 3, 3, 3, 4, 5 ] )
data = [ trace ]
py.iplot ( data )
```

2D Histogram

```
trace = go.Histogram2d (
    x = [ 1, 2, 3, 3, 3, 4, 5 ],
    y = [ 1, 2, 3, 3, 3, 4, 5 ] )
data = [ trace ]
py.iplot ( data )
```

MAPS

Bubble Map

```
trace = dict (
    type = 'scattergeo' ,
    lon = [ 100, 400 ], lat = [ 0, 0 ],
    marker = dict (
        marker = [ 'red', 'blue' ]
        size = [ 30, 50 ] ) ,
    mode = 'markers' )
py.iplot ( [ trace ] )
```

Choropleth Map

```
trc = dict (
    type = 'choropleth' ,
    locations = [ 'AZ', 'CA', 'VT' ],
    locationmode = 'USA-states' ,
    colorscale = [ 'Viridis' ],
    z = [ 10, 20, 40 ] )
lyt = dict ( geo = dict ( scope = 'usa' ) )
map = go.Figure ( data = [ trc ],
    layout = lyt )
py.iplot ( map )
```

Scatter Map

```
trace = dict (
    type = 'scattergeo' ,
    lon = [ 42, 39 ], lat = [ 12, 22 ],
    marker = [ 'Rome', 'Greece' ],
    mode = 'markers' )
py.iplot ( [ trace ] )
```

3D CHARTS

3D Surface Plots

```
trace = go.Surface (
    colorscale = 'Viridis' ,
    z = [ [ 3, 5, 8, 13 ],
          [ 21, 13, 8, 5 ] ] )
data = [ trace ]
py.iplot ( data )
```

3D Line Plots

```
trace = go.Scatter3D (
    x = [ 9, 8, 5, 1 ], y = [ 1, 2, 4, 8 ],
    z = [ 11, 8, 15, 3 ],
    mode = 'lines' )
data = [ trace ]
py.iplot ( data )
```

3D Scatter Plots

```
trace = go.Scatter3D (
    x = [ 9, 8, 5, 1 ], y = [ 1, 2, 4, 8 ],
    z = [ 11, 8, 15, 3 ],
    mode = 'markers' )
data = [ trace ]
py.iplot ( data )
```

FIGURE HIERARCHY

Figure { }

```
DATA [ ]
    TRACE { }
        x, y, z [ ]
        color, text, size [ ]
        colorscale ABC or [ ]
    MARKER { }
        color ABC
        symbol ABC
    LINE { }
        color ABC
        width 123

LAYOUT { }
    title ABC
    XAXIS, YAXIS { }
    SCENE { }
        XAXIS, YAXIS, ZAXIS { }
    GEO { }
    LEGEND { }
    ANNOTATIONS { }
```

```
{ } = dictionary
[ ] = list
ABC = string
123 = number
```

Plotly Express cheat sheet

 [Open in Colab](#)



General

Install

```
pip install plotly
```

Import

```
import plotly.express as px
```

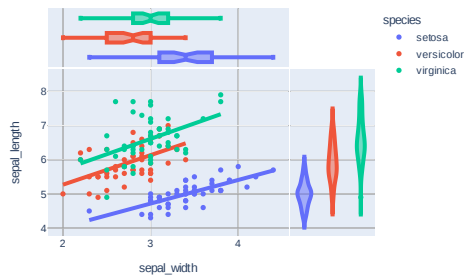
General usage

```
fig = px.chart_type(  
    df,  
    **chart_specific_parameters,  
    title="Chart title",  
    labels={"x_column_name": "X column name"},  
    width=600,  
    height=400,  
)  
fig.show()
```

Chart types

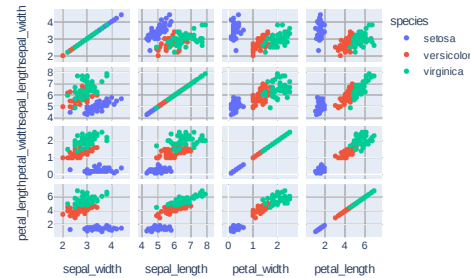
Scatter

```
px.scatter(  
    df,  
    x="sepal_width",  
    y="sepal_length",  
    color="species",  
    marginal_y="violin",  
    marginal_x="box",  
    trendline="ols",  
    hover_data=["petal_length", "petal_width"],  
)
```



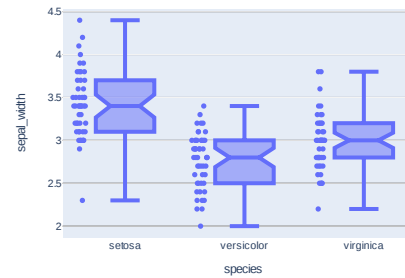
Scatter matrix

```
px.scatter_matrix(  
    df,  
    dimensions=[  
        "sepal_width",  
        "sepal_length",  
        "petal_width",  
        "petal_length",  
    ],  
    color="species",  
)
```



Box

```
px.box(  
    df,  
    x="species",  
    y="sepal_width",  
    notched=True,  
    points="all"  
)
```



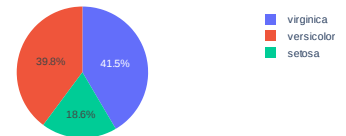
Bar

```
px.bar(count_df, x="species", y="count")
```



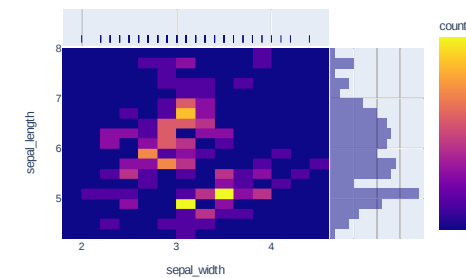
Pie

```
px.pie(count_df,  
    names="species", values="count")
```



Density heatmap (2D Histogram)

```
px.density_heatmap(  
    df,  
    x="sepal_width",  
    y="sepal_length",  
    nbinsx=20,  
    nbinsy=20,  
    histfunc="count", # or "sum", "avg"  
    marginal_x="rug",  
    marginal_y="histogram",  
)
```



More options

Facet plots

```
px.density_heatmap(  
    df,  
    x="sepal_width",  
    y="sepal_length",  
    facet_row="species",  
)
```

